

Smart Garden with IoT Based Real Time Communication using MQTT Protocol

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Abstract—Watering plants is a routine and most important task for plants to keep growing and developing. The automatic watering system can ease the human burden. Automation can be used or leveraged to help do the work at hand. This paper proposes an automatic smart garden watering system with parameters measuring temperature and humidity, and soil moisture using a DHT11 sensor, a soil moisture sensor, a water pump, and an ESP8266 connected to the internet network with optimization of data communication using the MQTT (Message Queue Telemetry Transport) protocol. This smart garden data monitoring system is carried out in real time continuously so that users can monitor anytime and anywhere via mobile devices. Each temperature and humidity sensor has an average error of 0.03%. Where the difference in the average temperature measurement is 0.054°C with a standard deviation value of 0.326, and the average difference in air humidity measurements is 0.03% with a standard deviation value of 0.30. A low-cost system designed using the MQTT protocol which is lightweight yet powerful enough to handle multiple devices at once.

Keywords—Smart Garden, MQTT, IoT, ESP8266

I. INTRODUCTION

Current technological progress is something that cannot be avoided and its progress will continue to go hand in hand with the development of science [1]. The innovations created are also beneficial for human needs which can even help human work in daily life. One of them is a man who loves plants so much that he has a private garden placed at home. Along with the high mobility of humans in this modern era, the need to facilitate personal garden maintenance requires a device and system that can care for and water plants automatically. The use of automatic technology has been so advanced that the use of daily activities can be done automatically because humans will not always use conventional methods. There have been many studies that have developed automation systems to be able to help humans do these routine things using Internet of Things (IoT) technology [2].

Currently, internet access requires TCP/IP and the Hyper Text Transfer Protocol (HTTP) application protocol. Anytime information is accessed via the HTTP Protocol, three-way communication is required for the Handshake process with TCP. When the HTTP protocol is applied to the IoT system, it will cause several problems in the IoT communication

system, namely: failure to send data, robustness of the system in real time, and efficiency of data transmission [3].

Many research on monitoring and control systems have been carried out. In [4], the authors built a controller that controls a small-scale greenhouse. The controller regulates multiple inputs and outputs of greenhouse temperature and humidity within a predefined range. The communication network architecture used is a GSM network with the MQTT protocol controlled with an Arduino DUE. In [5], the author designed a temperature and IR sensor with a Raspberry Pi 3 controller optimized with the MQTT protocol and QoS parameters. In [6], the author proposes smart greenhouse automation with IoT technology that uses a cellular GSM module. Data from the soil moisture sensor is sent via an internet connection using a 3G/4G gateway. The results of this study can save the use of unnecessary fertilizers. In [7], the author designed an analysis of the data communication rate in the IoT architecture using the MQTT protocol with a QoS approach.

This paper proposes a smart garden system using IoT technology based on Node MCU with optimization using the MQTT (Message Queue Telemetry Transport) protocol. This protocol has light weighted message properties, can work with low bandwidth, high latency and runs at the application layer. The MQTT protocol mechanism is publish/subscribe where the sending and receiving of MQTT protocol messages is based on predetermined topics [8]. In its implementation, it uses a small computer called a microcontroller. With the use of modern technology, it is hoped that it can increase much better results and maximize time efficiency so that owners can also monitor and find out information on existing measurement parameters such as temperature, humidity, soil moisture and watering that can be accessed remotely in real time.

II. METHODOLOGY

A. Smart Garden Design

Smart garden means a smart garden or garden, which can be interpreted as a system design made to facilitate work in the plantation sector. The smart garden has an automatic function and is equipped with several sensors for the monitoring process [9][10]. This section describes the scheme

of monitoring agricultural data sensor by adopting the prototyping process, Fig. 1.

The assembly process begins by initiating the process of a two-parameter management scheme, the condition of the plantation and plantation irrigation system. Then, in terms of hardware, software, and data transfer, establish the material needs. Once the material is available, a prototype is created in accordance with the design and must be inspected directly by the gardener, so that if an evaluation is made, an improvement process may be initiated. The gadget was then trialed until the gardener approved, at which point it was modified to his needs. To begin creating smart garden devices based on the internet of things, a communication process from the data flow must be carried out so that the essential tool needs are established.

This system works with 3 main blocks. The first block is the input, which consists of sensors of air temperature, humidity, and soil moisture sensors. The second part block is the output, which consists only of the relay used to drive the water pump. while the third block is the wireless communication block, which uses the MCU Node which is connected to the internet network via Wi-Fi communication. The Taman Pintar monitoring system uses the Adafruit MQTT application, so that planters can monitor environmental conditions around their plants.

B. Hardware and Network Requirement

The hardware needed in this research is divided into two parts, namely the hardware used to make internet of things (IoT) equipment, and the devices used to support communication and data storage circuits. Some of the main equipment are as follows; YL-69 soil moisture sensor, DHT11 temperature, and humidity sensor, ESP8266, Relay, 9V micro water pump, and acrylic case. To support data communication, hardware is needed of an access point for two-way communication installed on Wi-Fi devices placed in the field, and communication devices owned by gardeners or users [11][12].

The DHT11 temperature and humidity sensor is a low cost and reliable sensor [13][14]. It has 4 pins, where pin 1 is Vcc, pin 2 is a pin for data collected from the surrounding environment and channeled to the processor. The pin configuration of the DHT11 temperature and humidity sensor is shown in Figure 2a. This sensor also has high reliability and excellent long-term stability [15][16]. The DHT11 sensor has a resistive type temperature measurement component and is connected to an 8-bit microcontroller, offering high quality and fast response, anti-interference capability and an

affordable price. The temperature range is 0°C-55°C and the humidity range is 20-90% [17].

The YL69 soil moisture sensor is known as a hygrometer sensor to detect the level of soil moisture. To determine the moisture level of the soil, this sensor measures the resistance of the soil [18]. With two probes at the end of the sensor, the soil moisture sensor can determine how much water is in the soil. There is a module in a set of YL-69 type moisture sensors that has an IC LM393 that performs a low offset comparison procedure of less than 5mV that is highly steady and exact.

The detecting sensitivity may be changed by twisting the potentiometer integrated into the processing module. When utilizing a microcontroller or Arduino, you may utilize an analog output (connected to the ADC pin or analog input on the microcontroller) to deliver a humidity value on a scale of 0V (relative to GND) to VCC (power supply voltage). Figure 2b depicts a soil moisture sensor of type YL-69 [19][20].

The Node MCU is an Internet of Things (IoT) product development board based on the eLua Firmware and the ESP8266-12E System on a Chip (SoC). The ESP8266 is a WiFi chip that has a full TCP/IP protocol stack. [21]. Node MCU can be analogous to the Arduino ESP8266 board. The display of the ESP862 MCU node can be seen in Figure 3.

For the working voltage, the ESP8266 uses the JEDEC voltage standard (3.3V voltage) to function. Unlike the AVR microcontroller and most Arduino boards which have a TTL voltage of 5 volts [22][23]. Node MCU can also still be



Fig. 2. (a)DHT11 Sensor (b)YL-69 Sensor



Fig. 3. Node MCU ESP8266

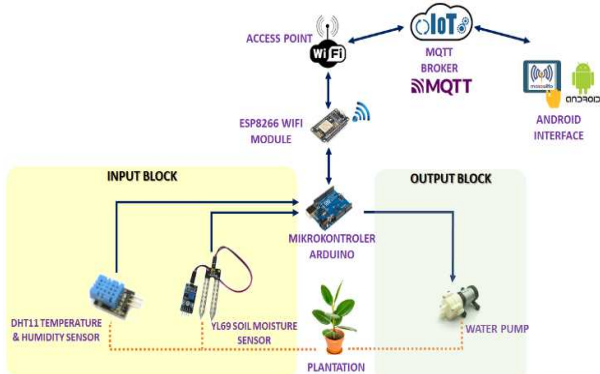


Fig. 1. IoT-based smart garden design architecture

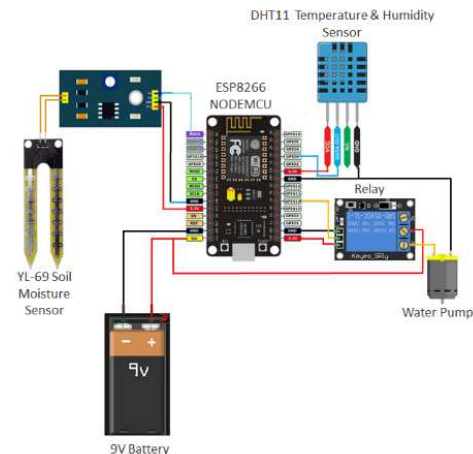


Fig. 4. The scheme of microcontrollers of Smart Garden

connected with 5V voltage but through the micro USB port or Vin pin provided by the board [21].

The following is a schematic of an IoT-based smart garden device using sensors of air temperature, humidity, and soil moisture.

C. MQTT Protocol

MQTT (Message Queuing Telemetry Transport) is an IoT connectivity protocol applicable to the use of small, low-power sensors. MQTT runs on top of the Transmission Control Protocol/Internet Protocol (TCP/IP) [24]. This protocol has low bandwidth usage and is usually operated in remote areas Sasaki [3].

The data is published by the Node MCU to MQTT Broker. Furthermore, MQTT Broker transmits data through the topic subscription process by the user. MQTT itself is integrated with the Internet of Things [25].

This system is designed using io.adafruit.com server as data monitoring server. Temperature, humidity, soil moisture, and output messages are published to the server with the topics Temperature, Humidity, SoilMoisture, and Output of Pump, respectively. The publishing implementation of this protocol on the io.adafruit.com server uses the Adafruit_MQTT_Publish command, which can be written as follows.

```
Adafruit_MQTT_Publish temperature =
Adafruit_MQTT_Publish(&mqtt,
AIO_USERNAME "/feeds/temp");
```

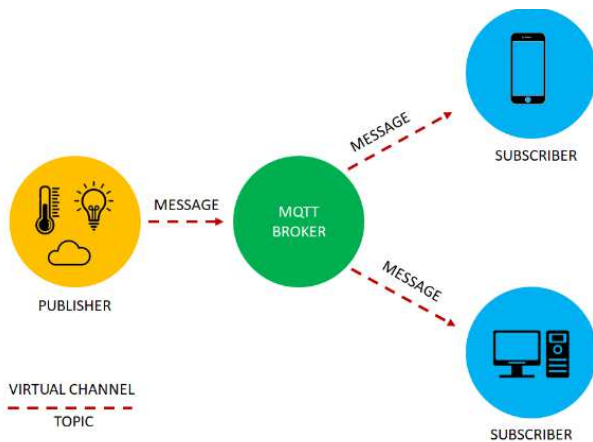


Fig. 5. MQTT block diagram [12].

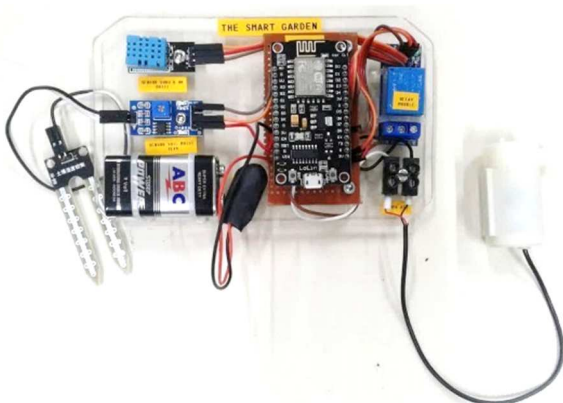


Fig. 6. IoT based smart garden prototype

```
Adafruit_MQTT_Publish humidity =
Adafruit_MQTT_Publish(&mqtt,
AIO_USERNAME "/feeds/humidity");
```

```
Adafruit_MQTT_Publish soilmoisture =
Adafruit_MQTT_Publish(&mqtt,
AIO_USERNAME "/feeds/soilmisture");
```

```
Adafruit_MQTT_Subscribe pump =
Adafruit_MQTT_Subscribe(&mqtt,
AIO_USERNAME "/feeds/pump");
```

III. RESULTS AND DISCUSSION

This Smart Garden device is installed in the BROTANI plantation in Malang City, with a total of 144 plants in a 3m² garden area. Figure 6 is a prototype of a smart garden device.

To examine the sensors of the smart device are working correctly, we calibrate the device using the Beamex type MC5 calibrator and the PT-100 standard (Platinum RTD) which is certified and traceable to the Indonesian Metrology Agency. The calibration process can be seen in Figure 7.

This Smart Garden gadget is placed in the midst of all planting material, according to study done by [26]. There are two types of measurement values provided in this study: measured values and datalogger values. The previous value is obtained from the data server io.adafruit.com. The value of the sensor measurement is subsequently relayed to the data server. The datalogger, which is installed on the plantation, provides the final value. Air temperature and humidity may be measured using the data logger's capabilities. These numbers serve as a check against the preceding ones. The findings of the temperature sensor measurement (DHT11) during a 24-hour period are shown in Figure 8.



Fig. 7. Smart Garden Calibration

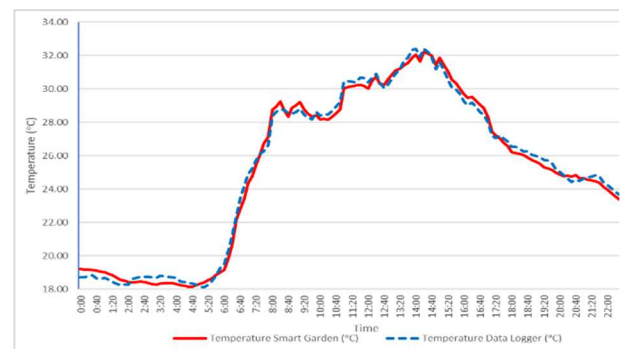


Fig. 8. Smart garden air temperature measurement compared with the datalogger value.

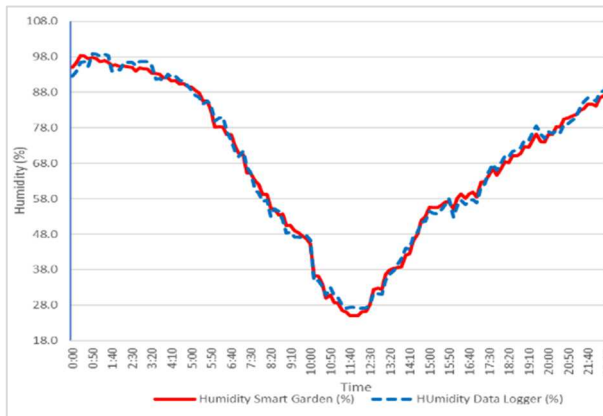


Fig. 9. Smart garden air humidity measurement compared with the datalogger value.

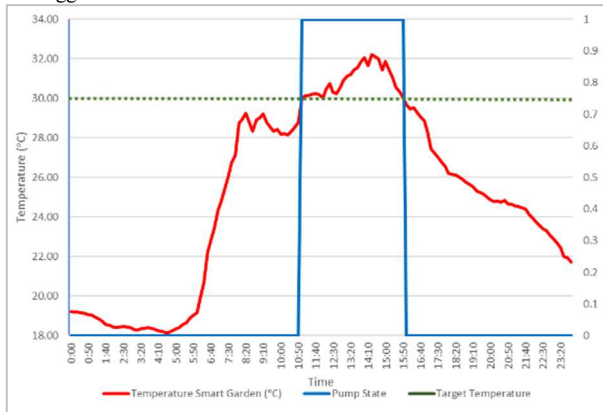


Fig. 10. Water pump output status against sensor measurement.

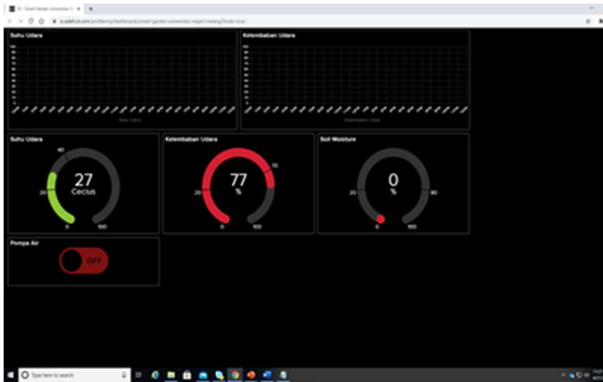


Fig. 11. Dashboard of measurement results at io.adafruit.com server

In Figures 8 and 9, the red line shows the measurement results of the smart device, while the blue dashed line shows the value of the data logger. The value of the measurement results of the smart garden device is not exactly the same as the datalogger measurement, but is almost identical. In general, the shape of the two lines for 24-hour measurements is almost the same. The difference in the average measurement of temperature is 0.054°C with 0.326 standard deviation value, and the difference in the average measurement of air humidity is 0.03% with 0.30 standard deviation value. This graph shows that the temperature and humidity sensor is working as expected. Based on data recorded 24 hours a day and repeated for 5 consecutive days, the average value of the difference in temperature measurement is 5.35% and air humidity measurement is -0.03%. Figure 10 shows a graph of the activation state of the pump output, which is triggered by air

temperature. When the measured air temperature value is higher than the reference value, the Output value will be "High" or 1. The output value 1 (High) will run the relay which then runs the pump to spray water to lower the temperature of the plants and increase the soil moisture of the plantation crops.

Figure 8 indicates that the air temperature is greater than the reference value from 10:50 to 15:50, i.e. we set 30°C, thus the water pump's output value is set to 1; otherwise, the value is 0. (zero). The pump will cease spraying water if the temperature falls below the standard threshold. The controller output has a value of 0.00 in the following 15 minutes, i.e. at 15:50, indicating that the engineering temperature is less than 30°C.

The MQTT protocol is used in the system developed as a result of our research. The MQTT protocol is used to send the measured data through the communication domain. Figure 11 illustrates the data dashboard for utilizing the procedure to measure plant conditions, such as temperature and humidity, which have values of 0.0540°C and 77 percent, respectively. Over WiFi connectivity, data is transmitted utilizing the MQTT protocol, which is lightweight and suited for IoT design. The data is subsequently stored on a data server, which acts as a middleman in allowing the measured data to be released in real time.

The smart garden system's capacity to send data in real time is quite high, which may improve the efficacy of monitoring and managing plants. This system is inexpensive to build and implement. The planters gained confidence in the state of their harvests. MQTT may be used in smart garden systems as well as bigger farm monitoring systems that require sensors in many locations.

IV. CONCLUSION

In this paper, we have presented the design, workflow, and implementation of an internet of things-based smart garden using the MQTT protocol. We focus on discussing the implementation of making smart gardens to identify plantation data in real time. The temperature and air humidity sensors have average errors by the value of 0.03%, respectively. The sensor data parameters are displayed in a web page so that it can be accessed on various devices that have a web browser. The system designed is low cost as it uses the MQTT protocol which is lightweight but powerful enough to handle multiple devices at once.

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